

Growth Predictor

Quantitative Stock Market Asset Analysis Model

Model Card V1.3

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Document Purpose

The following document serves as a report on the performance of the Growth Predictor model, and as a guide to its structure, intended use, and possible future improvements. Sections of this report include basic model information, training data structure, evaluation process, results, known limitations, and possible improvements.

Model Information

The Growth Predictor model forecasts the Compounded Annual Growth Rate of an asset by 1 year when given the past 5 years of an asset's price data. This is an aggregation model, which uses historical price data converted to CAGR values, and forecasts based on trained "intuition". Typically, share price or cash flow forecasting models like DCF or P/FCF and etc. use a growth rate as the core of their calculations since it scales future values. The growth rate most times is a projection of future growth inferred from historical CAGRs or price movements, and is an assumption most analysts build an "intuition" for, this ML model however aims to bridge that intuition gap.

$$\text{CAGR} = \left(\frac{\text{Ending Value}}{\text{Beginning Value}} \right)^{\frac{1}{n}} - 1$$

Figure 1: Compounded Annual Growth Rate formula

The output of the model will reflect the future growth rate a typical stock analyst may estimate, provided in a decimal (float) format, representing the CAGR estimation for the next year.

Key Model Dependencies

1. Joblib - for loading the serialized model artifact file
2. Pandas - when handling data matrices and array transformations
3. Scikit-learn (maybe) - loading x_model and StandardScaler

Model Input & Output

Input: An array of five float numbers, representing the Compounded Annual Growth Rates of an asset for the 5 years prior to the date of forecasting. At index [0] place CAGR_1 representing the first calculated growth between years 1 and 2, then at index [3] place CAGR_4 representing the last calculated growth between the years 4 and 5.

Output: A single float number representing the CAGR between current and next year, projecting next year's growth, and can be directly plugged into other models.

Model Architecture

To Do.

Training Dataset Sources

To Do.

Training Data Structure

To Do.